

Exponential Functions

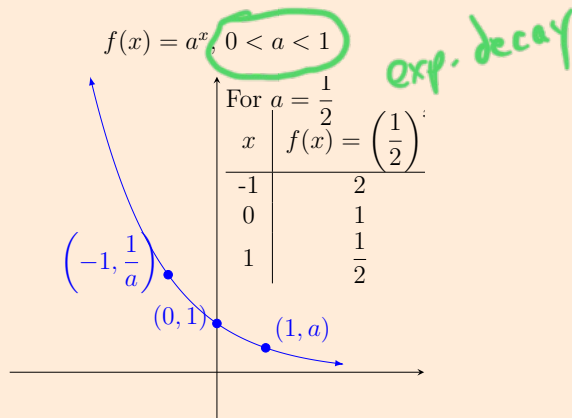
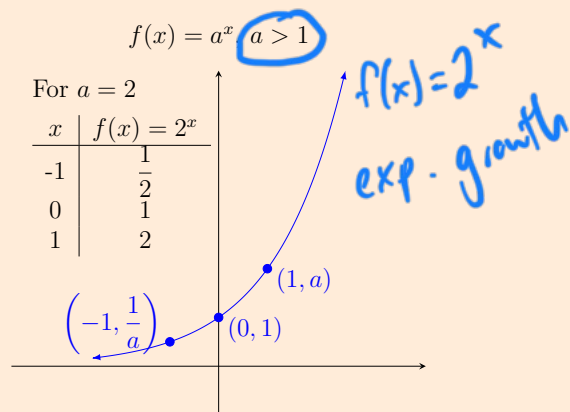
Exponential Function

If $a > 0$ and $a \neq 1$, then the exponential function with base a is

$$f(x) = a^x.$$

Characteristics of the Graph of $f(x) = a^x$

1. The point $(-1, \frac{1}{a})$, $(0, 1)$ and $(1, a)$ are on the graph.
2. If $a > 1$, then f is an increasing function. If $0 < a < 1$, then f is a decreasing function.
3. The x -axis is a horizontal asymptote.
4. The domain is $(-\infty, \infty)$ and the range is $(0, \infty)$.



Problem 8. * The function f is defined by the following rule $f(x) = 7^x$. Find $f(x)$ for each x -value in the table. Then graph f .

x	y	(x, y)
-3		
-2		
-1		
0		
1		

Problem 9. * The function f is defined by the following rule $f(x) = \left(\frac{1}{8}\right)^x$. Find $f(x)$ for each x -value in the table. Then graph f .

x	y	(x, y)
-2		
-1		
0		
1		
2		

Problem 10. * Graph the exponential function $f(x) = 3^x$. Plot five points on the graph of the function, and also draw the asymptote.

x	y	(x, y)
-2	$\frac{1}{9}$	$(-2, \frac{1}{9})$
-1	$\frac{1}{3}$	$(-1, \frac{1}{3})$
0	1	$(0, 1)$
1	3	$(1, 3)$
2	9	$(2, 9)$

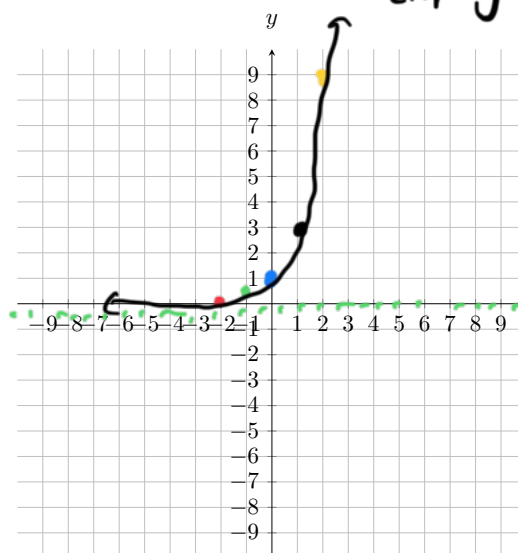
$$3^{-2} = \frac{1}{3^2} = \frac{1}{9}$$

$$3^{-1} = \frac{1}{3^1} = \frac{1}{3}$$

$$3^0 = 1$$

$$3^1 = 3$$

$$3^2 = 9$$



exp. growth
 $3 > 1$

Problem 11. * Graph the exponential function $f(x) = \left(\frac{1}{2}\right)^x$. Plot five points on the graph of the function, and also draw the asymptote.

x	y	(x, y)
-2	4	$(-2, 4)$
-1	2	$(-1, 2)$
0	1	$(0, 1)$
1	$\frac{1}{2}$	$(1, \frac{1}{2})$
2	$\frac{1}{4}$	$(2, \frac{1}{4})$

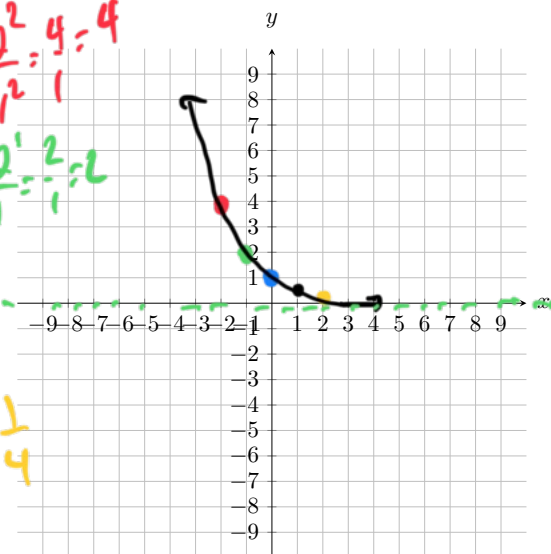
$$\left(\frac{1}{2}\right)^{-2} = \frac{1}{2^{-2}} = \frac{2^2}{1^2} = 4$$

$$\left(\frac{1}{2}\right)^{-1} = \frac{1}{2^{-1}} = \frac{2^1}{1^1} = 2$$

$$\left(\frac{1}{2}\right)^0 = 1$$

$$\left(\frac{1}{2}\right)^1 = \frac{1}{2}$$

$$\left(\frac{1}{2}\right)^2 = \frac{1^2}{2^2} = \frac{1}{4}$$



$0 < \frac{1}{2} < 1$ exp. decay

$y = 0$